

Problem 8 – Smart Movie Ticket Booking System

Difficulty: ★★ Beginner

Overview

A local cinema wants to replace its paper ticket booking process with a smart booking assistant. Customers should be able to select a movie, choose a show time, book tickets, and receive a booking summary with the total cost.

Your task is to build a Smart Movie Ticket Booking System using Streamlit.

Project Requirements

1. Store Cinema Information

Store the available movies, ticket prices, and show times using appropriate Python data structure(s).

Movies

Movie	Ticket Price (EGP)
Inside Out 2	120
Minecraft	150
Sonic 3	130
Kung Fu Panda 4	140

Show Times

- 10:00 AM
- 1:00 PM
- 4:00 PM
- 7:00 PM

Note: Choose the most appropriate Python data structure(s) to store this information.

2. Build the User Interface

Use Streamlit to create the application.

Required Text Functions

Use all of the following:

- `st.title()`
 - `st.header()`
 - `st.subheader()`
 - `st.write()`
 - `st.divider()`
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3. Create the Booking Form

Allow the customer to enter:

- Customer Name
- Movie
- Show Time
- Number of Tickets
- Add Popcorn + Soft Drink Combo (Yes/No)

Use the following widgets.

Widget	Purpose
<code>st.text_input()</code>	Customer Name
<code>st.selectbox()</code>	Movie
<code>st.selectbox()</code>	Show Time
<code>st.number_input()</code>	Number of Tickets
<code>st.selectbox()</code>	Add Combo
<code>st.button()</code>	Confirm Booking

4. Validate User Input

Before confirming the booking, ensure that:

- Customer name is not empty.
- Number of tickets is greater than zero.

Display an appropriate error message if the booking information is invalid.

5. Calculate the Total Cost

Calculate the booking total.

Total Cost = (Ticket Price × Number of Tickets) + Combo Price (if selected)

Combo Price = 60 EGP

6. Save the Booking

When Confirm Booking is clicked:

- Validate the booking.
 - Calculate the total cost.
 - Save the booking record.
 - Display a success message.
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7. Display Movie Schedule

Create a page that displays:

- Movie Name
 - Ticket Price
 - Available Show Times
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8. Display Booking Summary

After a booking is successfully confirmed, display a **Booking Summary** showing:

- Customer Name
- Movie
- Show Time
- Number of Tickets
- Combo (Added / Not Added)
- Total Cost

Example

Booking Summary

Customer : Omar

Movie : Sonic 3

Show Time : 4:00 PM

Tickets : 3

Combo : Added

Total : 450 EGP

Bonus Mission ★

Allow customers to add a Popcorn + Soft Drink Combo for 60 EGP.

If selected:

- Add 60 EGP to the total bill.
 - Display that the combo has been added.
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Required Status Elements - Use at least four of the following.

Streamlit Function	Suggested Usage
st.info()	Display today's featured movie or a cinema announcement in the sidebar.
st.success()	Booking confirmed successfully.
st.warning()	Customer books more than 5 tickets in a single booking.
st.error()	Invalid customer name or invalid number of tickets.

These uses naturally fit each status element.

Required Layout

Sidebar

Display:

- Movie List
- Ticket Prices
- Cinema Rules

Tabs

Create three tabs.

Tab 1 – Book Tickets

Contains the booking form.

Use two columns.

Left Column

- Customer Name
- Movie
- Show Time
- Number of Tickets
- Combo Selection
- Confirm Booking Button

Right Column

Display a Booking summary.

Tab 2 – Movie Schedule

Display all available movies.

For each movie, display:

- Movie Name
 - Ticket Price
 - Available Show Times
-

Tab 3 – Booking Report

Display all confirmed booking.

If no booking has been made yet, display an appropriate message.

Booking Report

Viewed in a table contains the following information

Customer Name

Movie

Show Time

Tickets

Combo

Total

Challenge Mission ★★

Create a Daily Cinema Report that displays:

- Total Number of Bookings
- Total Tickets Sold
- Total Sales (EGP)
- Most Booked Movie
- Most Popular Show Time

Store booking records using an appropriate data structure.

Useful Hints

- Choose suitable data structures to store movies and bookings.
 - Create reusable functions such as:
 - calculate_total()
 - save_booking()
 - display_schedule()
 - display_booking_summary()
 - Use st.session_state to keep booking records while the application is running.
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Expected GUI



Smart Movie Ticket Booking System

Sidebar

Movie List

Ticket Prices

Cinema Rules

Tabs

[Book Tickets] [Movie Schedule] [Booking Report]

Customer Name [Omar]

Movie [Sonic 3 ▼]

Show Time [4:00 PM ▼]

Tickets [3]

Combo [Yes ▼]

[Confirm Booking]

Booking Summary

Customer : Omar

Movie : Sonic 3

Show Time : 4:00 PM

Tickets : 3

Combo : Added

Total : 450 EGP



Booking Confirmed!

Python Topics Covered

- Variables
 - Strings
 - Lists and/or Dictionaries
 - Integers
 - Arithmetic Operators
 - Conditional Statements
 - User-defined Functions
 - `st.session_state`
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Streamlit Topics Covered

Display

- `st.title()`
- `st.header()`
- `st.subheader()`
- `st.write()`
- `st.divider()`

Input Widgets

- `st.text_input()`
- `st.selectbox()`
- `st.number_input()`
- `st.button()`

Status Elements

- `st.info()`
- `st.success()`
- `st.warning()`
- `st.error()`

Layout

- `st.sidebar`
 - `st.tabs()`
 - `st.columns()`
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Learning Objectives

After completing this project, you should be able to:

- Choose appropriate Python data structures to represent application data.
- Validate user input before processing a booking.
- Calculate totals based on user selections.
- Store and display booking records.
- Organize a program into reusable functions.
- Build a multi-tab Streamlit application that separates booking, schedule viewing, and booking summaries into a clear and user-friendly interface.